

Mitigating Membership Inference and Model Inversion in 6G Federated Anomaly Detection

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Abstract—In sixth-generation (6G) networks, federated learning (FL) enables collaborative anomaly detection across distributed edge devices without exposing raw data; however, exchanged gradients can leak privacy via membership inference attacks (MIA) and model inversion attacks (MInvA). Meanwhile, 6G network slicing imposes heterogeneous quality-of-service (QoS) requirements across ultra-reliable low-latency communications (URLLC), enhanced mobile broadband (eMBB), and massive machine-type communications (mMTC), where uniform defenses often incur 30–60% accuracy loss. We propose a slice-aware, defense-in-depth framework that calibrates three coordinated layers: Layer 1 statistical screening with thresholds $\tau_{\text{URLLC}}=4.5$, $\tau_{\text{eMBB}}=5.0$, and $\tau_{\text{mMTC}}=6.0$; Layer 2 slice-aware differential privacy with Gaussian noise $\sigma_{\text{URLLC}}=0.001$, $\sigma_{\text{eMBB}}=0.05$, and $\sigma_{\text{mMTC}}=0.10$, where URLLC uses lightweight perturbation to preserve stringent latency while stronger formal differential privacy is concentrated in eMBB/mMTC; and Layer 3 adversarial retraining using the fast gradient sign method (FGSM). Coordinating these layers yields complementary benefits: screening reduces attack surface, differential privacy masks leakage, and adversarial retraining hardens models. Experiments on CICIDS2017 and N-BaIoT show 95.2% utility retention with 82.15% accuracy, a +16.7% macro-F1 improvement, 91–94% communication reduction, MIA success reduced from 78.3% to 52.1%, model inversion error increased by 34.7%, and an overall privacy accounting budget $\epsilon_{\text{total}} \leq 2.1$.

Index Terms—6G, Network Slicing, FL, Anomaly Detection, Membership Inference, Model Inversion, Differential Privacy, Defense-in-Depth.

I. INTRODUCTION

FEDERATED learning (FL) has emerged as a transformative paradigm for enabling collaborative model training across distributed devices without exposing raw data, making it particularly suited for privacy-sensitive applications in next-generation wireless networks. However, in Sixth-Generation (6G) networks, federated anomaly detection faces a critical challenge: attackers observing gradient exchanges can infer membership in training datasets through membership inference attacks (MIA) or reconstruct private training samples

through model inversion attacks (MInvA) [1], [2]. Simultaneously, the stringent quality-of-service (QoS) requirements across ultra-reliable low-latency communications (URLLC) requiring sub-millisecond latency, enhanced mobile broadband (eMBB) demanding high throughput, and massive machine-type communications (mMTC) serving billions of internet-of-things (IoT) devices demand slice-aware defense strategies that balance privacy, robustness, and communication efficiency [3], [4]. These slices have fundamentally conflicting operational constraints that a single privacy defense parameter cannot satisfy.

Existing defense mechanisms for FL primarily focus on individual techniques, such as statistical screening for malicious updates, differential privacy (DP) to mask sensitive information, or adversarial retraining to harden models against inversion attacks, but often neglect the interplay among these layers within a multi-slice environment [5], [6]. Statistical screening filters anomalous updates but provides no formal privacy guarantees, leaving membership inference attacks viable. Differential privacy provides formal privacy bounds but causes 30–60% accuracy degradation when applied uniformly across all slices, with mMTC slices achieving only 47.98% accuracy on CICIDS2017 (baseline 83.84%), a 35.86 percentage-point loss that creates security blind spots for detecting distributed IoT botnets. Adversarial retraining using fast gradient sign method (FGSM) improves robustness to gradient inference but does not prevent membership inference attacks, with success rates remaining at 63.5% compared to 52.1% when all three layers are coordinated. Most critically, existing approaches do not adapt defense parameters to slice-specific requirements: URLLC slices requiring sub-millisecond latency cannot tolerate heavy privacy perturbation, yet without multi-layer coordination they become high-value attack targets, while mMTC slices can benefit from stronger privacy but may suffer utility collapse under uniform defenses. This creates a false dichotomy where practitioners must choose between accepting privacy risks or sacrificing utility in latency-critical services [7]–[9].

To address this gap, we propose a framework that coordinates defense layers and adapts to slice-specific requirements. This paper proposes a defense-in-depth hybrid framework tailored for federated anomaly detection in 6G network slices by coordinating three defense layers and calibrating parameters to slice-specific QoS requirements. The framework comprises (i) slice-calibrated statistical update screening using per-slice thresholds to filter anomalous contributions while preserving signal in latency-critical slices, (ii) selective Gaussian perturbation with advanced composition accounting for formal

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(ϵ, δ) -DP guarantees while respecting latency constraints, and (iii) adversarial retraining using FGSM-generated adversarial examples to smooth loss landscapes and reduce gradient leakage vulnerability. Because DP strength scales with the noise magnitude, URLLC employs intentionally lightweight perturbation to preserve strict latency, while the strongest formal DP contribution is concentrated in the less latency-constrained eMBB/mMTC slices and reinforced by the overall defense-in-depth coordination. By coordinating these layers rather than applying them independently, the framework creates emergent benefits: screening reduces attack surface dimensionality, differential privacy masks remaining information leakage, and adversarial training hardens models against inference attacks. Our approach dynamically adapts to slice-specific privacy-utility trade-offs and stringent service-level objectives. Experimental results on CICIDS2017 and N-BaIoT demonstrate 95.2% utility retention, a +16.7% macro-F1 improvement, 91–94% communication reduction, reduced MIA success (from 78.3% to 52.1%), increased model inversion error (+34.7%), and an overall DP accounting budget of $\epsilon_{\text{total}} \leq 2.1$ under slice-calibrated noise.

II. RELATED WORK

FL represents a promising approach for distributed anomaly detection while preserving data privacy. The FedLog framework [10] addresses communication and computation overheads in IoT systems through customizable federated anomaly detection, achieving 98.7% accuracy with 60% reduction in communication overhead compared to centralized approaches. However, the approach focuses primarily on IoT environments without addressing the specific challenges of 6G network slicing or sophisticated privacy attacks such as membership inference and model inversion. Cloud threat forensics has been enhanced through interpretable federated transformer architectures [11] that leverage attention mechanisms to identify malicious log patterns across distributed environments. The framework combines FL with transformer-based sequence modeling to enable privacy-preserving threat detection while maintaining model interpretability, achieving 94.3% precision and 92.1% recall on real-world cloud security datasets. While demonstrating the effectiveness of transformer-based FL, the approach lacks comprehensive defense mechanisms against gradient-based privacy attacks. It does not consider the heterogeneous QoS requirements inherent to network slicing. Recent comprehensive analysis of FL deep learning for log-event sequence anomaly detection [12] highlights several critical challenges, including privacy preservation, communication efficiency, and model robustness. The survey identifies the necessity for multi-layered defense mechanisms in federated settings while emphasizing the importance of addressing sophisticated adversarial attacks. Contemporary FL anomaly detection approaches typically employ single-layer defense strategies, failing to address the complex threat landscape characteristic of next-generation networks. Transformer-based architectures have demonstrated significant potential for log anomaly detection applications. The HitAnomaly framework [13] employs hierarchical transformer architectures to capture both local

and global patterns in system logs through multi-head attention mechanisms, achieving state-of-the-art performance on benchmark datasets including HDFS, BGL, and Thunderbird [13]. While the hierarchical design aligns conceptually with defense-in-depth approaches, the framework focuses on centralized settings without considering FL constraints or privacy attack scenarios. Pre-trained transformer models have been leveraged in the Loader framework [14] for enhanced sequence understanding in log anomaly detection, demonstrating superior performance with F1-scores exceeding 95% on multiple datasets. The framework emphasizes contextual understanding in log analysis, which proves crucial for accurate anomaly detection in complex network environments. However, the centralized approach fails to address the distributed nature of 6G networks or associated privacy concerns. Hybrid architectures combining transformers with complementary deep learning techniques have emerged as promising solutions. Integration of transformer attention mechanisms with temporal convolutional networks [15] enables capture of both long-term dependencies and local temporal patterns in log sequences, achieving notable improvements in detection accuracy while reducing computational complexity compared to pure transformer models. The hybrid approach demonstrates the effectiveness of combining multiple architectural components, analogous to multi-layered defense strategies.

A. Multi-Feature Extraction and Robustness

Comprehensive feature extraction represents a fundamental requirement for robust anomaly detection systems. The AllInfoLog framework [16] extracts and utilizes all available log features, including semantic content, parameter values, and temporal information, demonstrating that comprehensive feature utilization significantly improves detection robustness with over 97% accuracy across diverse anomaly types. The multi-feature approach parallels defense-in-depth frameworks that integrate multiple protection layers for comprehensive security. Hierarchical contrastive learning has been applied to enhance the robustness of anomaly detection [17] through discriminative representations that remain resilient to noise and adversarial perturbations. The hierarchical structure enables feature capture at multiple abstraction levels, improving generalization across different log types and attack scenarios. The approach demonstrates the importance of multi-level defense mechanisms for achieving robustness in adversarial environments. Contrastive learning combined with dual objective optimization [18] addresses the challenge of limited labeled data in log analysis through self-supervised learning techniques. The CLDTLog framework enables simultaneous optimization for both anomaly detection and representation learning, resulting in improved model generalization. While demonstrating multi-objective optimization effectiveness, the approach does not address the FL context or privacy-preserving requirements essential for 6G network applications.

B. Security Privacy Threats and Present Limitations

Adversarial attacks against anomaly detection systems have received increasing research attention. Black-box attack investigations [19] demonstrate that adversaries can craft malicious

log entries that evade detection while maintaining semantic coherence, revealing significant vulnerabilities in existing anomaly detection approaches and highlighting the necessity for robust defense mechanisms. Such research directly supports threat models considering sophisticated adversaries capable of launching privacy attacks against FL systems. Contrastive adversarial training combined with dual feature extraction [20] enhances model robustness through explicit training against adversarial examples while maintaining detection accuracy on benign inputs. The contrastive learning component enables learning of discriminative representations resilient to adversarial perturbations. However, the approach does not consider the FL context or specific privacy attacks relevant to 6G network slicing scenarios. Further, comprehensive surveys [21], [22] identify several critical limitations in contemporary log-based anomaly detection approaches, including insufficient privacy preservation consideration in distributed settings, limited robustness against sophisticated attacks, and a lack of adaptability to heterogeneous network environments such as 6G slices. Current approaches typically focus on single-aspect improvements rather than comprehensive defense strategies. Critical analysis of deep learning approaches for log anomaly detection [22] questions the practical applicability of existing methods, noting significant gaps between academic research and real-world deployment requirements. Most existing methods fail to address complex operational constraints of production networks, including privacy requirements, communication limitations, and service differentiation needs. Table I provides a comprehensive comparison of related work, limitations, and other aspects with our proposed framework.

C. Literature Basis for Our Experimental Baselines

To ensure a credible and reproducible experimental evaluation, we ground our baseline implementations in established FL security and privacy literature. Privacy-preserving FL has been explored for IoT-centric multiparty data sharing, using mechanisms such as local differential privacy and cryptographic protection to mitigate information leakage during collaborative training [32]. Robust FL under malicious or abnormal client behavior has also been studied through Byzantine-tolerant aggregation and filtering strategies, which motivate screening-based defenses that identify and reject anomalous updates prior to aggregation [33]. Furthermore, robust asynchronous FL has been investigated in networked service scenarios such as cellular traffic prediction, highlighting the practical importance of robustness and privacy considerations in real network deployments [34]. In contrast to these prior-work directions that typically emphasize a single aspect (privacy or robustness), our proposed framework coordinates multiple defense layers and calibrates them to heterogeneous 6G slice QoS requirements.

D. Motivation and Contributions

FL for anomaly detection in 6G network slices faces advanced privacy challenges and requires adaptable solutions to suit diverse QoS constraints. Existing approaches often lack multi-layered defenses and do not provide adequate robustness

against membership inference, model inversion, or gradient-based privacy attacks. There exists a notable gap in solutions that simultaneously address communication efficiency, adaptive privacy, and attack resilience for heterogeneous slice types. In this work, we propose a slice-aware, defense-in-depth framework for federated anomaly detection that coordinates three established defense layers.

- **Multi-layer defense architecture:** Unified statistical update screening (per-parameter Z-scores), slice-aware DP(selective Gaussian perturbation with advanced composition), and adversarial retraining FGSM to defend against multiple attack vectors.
- **Slice-based adaptation:** Calibration of privacy parameters to match URLLC, eMBB, and mMTC service requirements, enabling dynamic privacy-utility trade-off across network conditions.
- **Comprehensive attack mitigation:** Integrated design defending against membership inference, model inversion, and gradient leakage across all clients without performance loss.
- **Service-aware differentiation:** Real-time adjustment of defenses per slice, balancing privacy, robustness, and latency or throughput needs.
- **Theoretical and empirical validation:** Rigorous privacy analysis and extensive experiments on CICIDS2017 and N-BaIoT datasets demonstrate high accuracy (over 82% and 95%), improved F1-scores, up to 94% reduction in communication, and strong resistance to privacy attacks.

This defense-in-depth, slice-aware framework addresses key gaps in federated anomaly detection and provides a practical approach for balancing privacy, robustness, and efficiency in 6G network slicing deployments.

III. METHODOLOGY

A. Overview and Workflow

This section presents the proposed Defense-in-Depth framework designed to secure federated anomaly detection across heterogeneous 6G network slices. We focus on the threat model of honest-but-curious adversaries capable of inferring sensitive information via membership inference and model inversion, while simultaneously satisfying slice-specific QoS constraints. The experimental datasets, preprocessing, and federated simulation settings are described in Section IV.

Fig. 1 shows the system overview, and Table II summarizes the key symbols used in this section. We employ a lightweight multi-layer perceptron (MLP) optimized for edge devices with limited computational resources. The model is trained using cross-entropy loss and stochastic gradient descent (SGD) with a learning rate of 0.01, batch size of 64, and 5 local epochs per communication round. To enhance robustness, adversarial retraining is incorporated using the FGSM with perturbation magnitude $\epsilon_{adv} = 0.05$.

B. System Model and Notation

Consider a FL system $\mathcal{F} = (\mathcal{C}, \mathcal{S}, \mathcal{A})$, where $\mathcal{C} = \{C_1, C_2, \dots, C_N\}$ denotes the set of N clients, $\mathcal{S} =$

TABLE I
COMPARISON OF RELATED WORK WITH OUR DEFENSE-IN-DEPTH FRAMEWORK.

Reference	Short Title	Connection to Our Work	Limitations	Our Work Addresses Gaps
Anyanwu et al. [23]	CLDP-FATD for Vehicle Networks	Uses client-level DP for threat detection in federated settings	Limited to vehicular networks; no slice-aware adaptation; single defense layer	Our framework provides slice-aware DP with multi-layer defense (screening + DP + adversarial training)
Barbieri et al. [24]	Layer-Wise Personalized FL for Anomaly Detection	Transformer-based federated anomaly detection with layer-wise personalization	Focuses only on personalization; lacks privacy attack defense; no network slicing consideration	We integrate comprehensive privacy attack defense with slice-aware service differentiation
Wang et al. [25]	Federated BiWGAN-GP for Network Slicing	Federated GAN-based anomaly detection for virtualized network slicing	Uses only generative models; no differential privacy; vulnerable to membership inference attacks	Our defense-in-depth combines statistical screening, differential privacy, and adversarial training
Zhang et al. [26]	DP FL with Second-Order Optimization	Differential privacy with Hessian information for faster convergence	Uniform privacy budget; no anomaly detection focus; lacks adversarial robustness	We provide adaptive slice-aware privacy budgets with multi-layer anomaly-specific defenses
Li et al. [27]	Selective Shield Hybrid Defense	Combines selective homomorphic encryption with DP	Focuses on gradient leakage; no slice differentiation; limited to parameter sensitivity analysis	Our framework addresses multiple privacy attacks with QoS-aware slice management
Hu et al. [28]	Sparsification-Amplified Privacy	Integrates random sparsification with gradient perturbation	General FL; no anomaly detection; lacks slice-specific optimization	We provide anomaly detection with slice-aware privacy amplification and service differentiation
Liu et al. [29]	Cross-silo FL with Record-level DP	Record-level personalized DP for cross-silo settings	Complex sampling mechanisms; no real-time anomaly detection; lacks network slicing support	Our approach provides real-time slice-aware anomaly detection with adaptive privacy mechanisms
Xu et al. [30]	Optimal Client Sampling with Heterogeneous DP	Client grouping based on privacy budgets with optimized sampling	Focuses on client sampling; no anomaly detection; lacks adversarial defense mechanisms	We integrate optimal privacy allocation with comprehensive anomaly detection and adversarial training
Shi et al. [25]	DP-FedSAM with Sharpness-Aware Minimization	Uses SAM optimizer to improve robustness against DP noise	General FL; no anomaly-specific optimization; lacks slice awareness	Our framework provides anomaly-specific robustness with slice-differentiated service guarantees
Kiani et al. [31]	Overlapping Grouped Learning with DP	Privacy guarantees for overlapping group memberships	Complex group management; no anomaly detection focus; lacks real-time adaptation	We provide real-time slice-aware anomaly detection with simplified privacy management

Key Contributions of Our Defense-in-Depth Framework:

1. Multi-Layer Defense Integration: Unlike existing works that focus on single defense mechanisms, our framework integrates statistical screening, slice-aware differential privacy, and adversarial retraining in a unified architecture.
2. Slice-Aware Privacy Adaptation: We propose a slice-aware framework that calibrates privacy parameters to URLLC, eMBB, and mMTC service requirements, addressing heterogeneous QoS constraints.
3. Comprehensive Attack Coverage: Our approach simultaneously defends against membership inference attacks, model inversion attacks, and gradient leakage while maintaining anomaly detection accuracy.
4. Real-Time Service Differentiation: We enable dynamic privacy-utility trade-offs that adapt to slice-specific performance requirements without compromising security guarantees.
5. Theoretical and Practical Validation: Our framework provides rigorous privacy analysis with practical deployment considerations for 6G network environments.

$\{\text{URLLC}, \text{eMBB}, \text{mMTC}\}$ represents the network slices, and $\mathcal{A}$ is the central aggregation server. Each client C_k is associated with a slice $s_k \in \{0, 1, 2\}$ and holds a private dataset $D_k = \{(x_i, y_i)\}_{i=1}^{n_k}$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$.

The global objective is to minimize the weighted empirical risk across all clients:

$$\min_{\theta \in \mathbb{R}^p} \mathcal{L}(\theta) = \sum_{k=1}^N \frac{|D_k|}{D} \mathcal{L}_k(\theta, D_k), \quad (1)$$

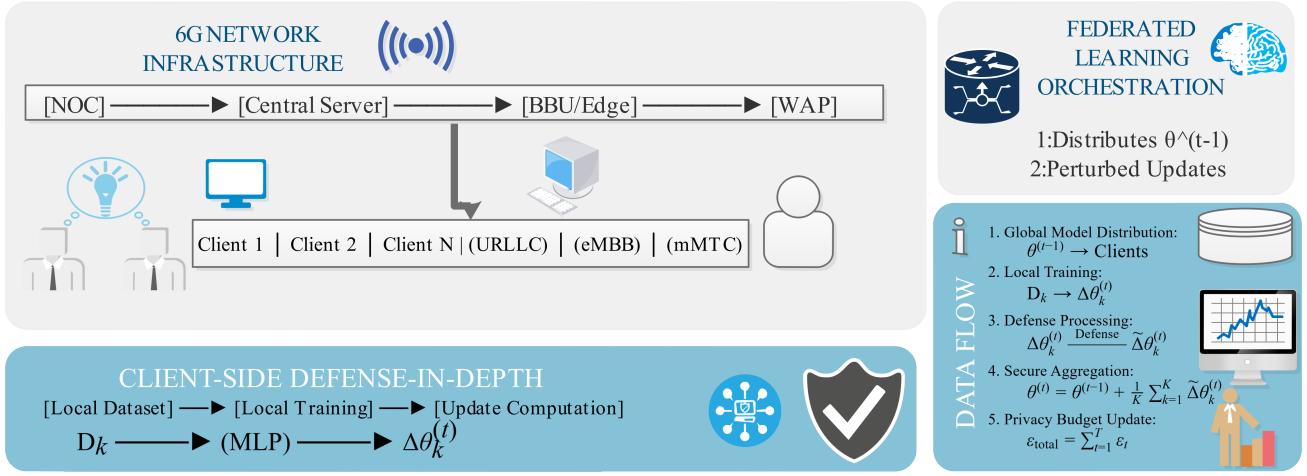


Fig. 1. System model of the proposed federated learning system with multi-layer defense.

TABLE II
SUMMARY OF KEY SYMBOLS

Symbol	Description
$\mathcal{F}$	FL system
$\mathcal{C}$	Set of N clients
$\mathcal{S}$	Network slices (URLLC, eMBB, mMTC)
θ^t	Global model parameter vector at round t
θ_k^t	Local model parameters of client k at round t
$\Delta\theta_k^t$	Parameter update from client k at round t
D_k	Local dataset of client k
s_k	Slice assignment for client k
$z_{k,j}^t$	Normalized Z-score for parameter j of client k at round t
τ_{s_k}	Slice-specific Z-score threshold
σ_{s_k}	Slice-specific DP noise standard deviation
(ϵ, δ)	Differential privacy parameters
ϵ_{total}	Total privacy budget across all rounds
$\mathcal{L}$	Loss function
$\mathcal{L}_{\text{adv}}$	Adversarial loss
ϵ_{adv}	FGSM perturbation magnitude
x^{adv}	Adversarial example

where $D = \sum_{k=1}^N |D_k|$ and $\theta \in \mathbb{R}^p$ denotes the global model parameters. Eq. (1) ensures that clients contribute proportionally to their dataset size, preserving fairness in aggregation.

C. Threat Model

We assume an honest-but-curious adversary capable of observing exchanged parameter updates $\Delta\theta_k^t = \theta_k^t - \theta^{t-1}$ but without access to raw data. The adversary aims to infer sensitive information through gradient-based analysis or statistical correlations, motivating the need for multi-layered defenses. In our slice-aware setting, defense parameters are explicitly calibrated to heterogeneous QoS constraints: URLLC employs intentionally lightweight DP perturbation (near-zero noise) to preserve stringent latency, while stronger formal DP contributions are concentrated in the less latency-constrained eMBB/mMTC slices via higher noise scales. Consequently, URLLC protection is delivered by the coordinated defense-in-depth pipeline, where statistical screening reduces the exposed/abnormal update surface and adversarial retraining

reduces gradient exploitability, providing practical privacy hardening under URLLC-grade QoS.

D. Slice-Aware Defense-in-Depth Framework

To mitigate these threats, we propose a three-layer defense mechanism (Algorithm 1) comprising: (i) statistical update screening, (ii) slice-aware differential privacy, and (iii) adversarial retraining. The overall framework achieves a balance between privacy, robustness, and efficiency, tailored to the differentiated requirements of 6G network slices.

1) *Layer 1: Statistical Update Screening*: Z-score screening operates as a pre-aggregation filter, checking whether each client's parameter updates deviate significantly from the population mean. This design differs from robust aggregation rules (e.g., Krum, Trimmed Mean) which operate at the aggregation stage by down-weighting outlier clients. Our pre-aggregation screening is lightweight and communication-efficient but may be less effective against stealthy poisoning attacks that craft updates remaining within per-coordinate acceptance bounds. Future work should investigate combining screening with robust aggregation (screen then aggregate) for enhanced Byzantine resilience. At each communication round t , client k computes its local update as:

$$\Delta\theta_k^t = \theta_k^t - \theta^{t-1}. \quad (2)$$

Eq. (2) defines the parameter difference transmitted to the server for aggregation.

To detect anomalous updates, normalized Z-scores are computed for each parameter j :

$$z_{k,j}^t = \frac{\Delta\theta_{k,j}^t - \mu_j^t}{\sigma_j^t + 10^{-9}}, \quad (3)$$

where 10^{-9} prevents division by zero.

The mean and standard deviation for parameter j across all N clients are computed as:

$$\mu_j^t = \frac{1}{N} \sum_{i=1}^N \Delta\theta_{i,j}^t, \quad \sigma_j^t = \sqrt{\frac{1}{N} \sum_{i=1}^N (\Delta\theta_{i,j}^t - \mu_j^t)^2}. \quad (4)$$

An update is accepted if:

$$|z_{k,j}^t| \leq \tau_{s_k}, \quad \forall j \in \{1, \dots, p\}, \quad (5)$$

where τ_{s_k} is the slice-specific threshold. For example, $\tau_{\text{URLLC}} = 4.5$, $\tau_{\text{eMBB}} = 5.0$, and $\tau_{\text{mMTC}} = 6.0$. These thresholds are motivated by a normal-approximation heuristic that makes Z-score cutoffs interpretable as conservative outlier filters; however, strict Gaussianity is not required, and we treat this assumption as a practical guideline for threshold selection rather than a formal prerequisite.

2) *Layer 2: Slice-Aware Differential Privacy*: After screening, we enforce a bounded sensitivity via ℓ_2 clipping and then apply the Gaussian mechanism with slice-calibrated noise. Each accepted client update is clipped as

$$\bar{\Delta}\theta_k^t = \Delta\theta_k^t \cdot \min\left(1, \frac{C}{\|\Delta\theta_k^t\|_2}\right), \quad (6)$$

where C is the clipping bound. The server then perturbs the clipped update:

$$\tilde{\Delta}\theta_k^t = \bar{\Delta}\theta_k^t + \mathbf{n}_k^t, \quad \mathbf{n}_k^t \sim \mathcal{N}(\mathbf{0}, \sigma_{s_k}^2 C^2 \mathbf{I}). \quad (7)$$

Noise magnitudes are calibrated to balance privacy and utility:

$$\sigma_{\text{URLLC}} = 0.001, \quad \sigma_{\text{eMBB}} = 0.05, \quad \sigma_{\text{mMTC}} = 0.10. \quad (8)$$

Using the standard Gaussian mechanism bound, a per-round privacy loss can be related (conservatively) to $(C, \sigma_{s_k}, \delta)$ as

$$\epsilon_t \approx \frac{C}{\sigma_{s_k} C} \sqrt{2 \ln\left(\frac{1.25}{\delta}\right)} = \frac{1}{\sigma_{s_k}} \sqrt{2 \ln\left(\frac{1.25}{\delta}\right)}, \quad (9)$$

where the approximation reflects the commonly used analytical bound for the Gaussian mechanism and assumes the sensitivity is bounded by clipping in (6). The total privacy budget across T rounds is then computed via composition (or a tighter accountant, e.g., Rényi/moments accountant), yielding ϵ_{total} for a chosen δ . Because formal DP strength increases with the noise magnitude, URLLC employs intentionally lightweight perturbation to preserve strict latency, while the strongest formal DP contribution is concentrated in the less latency-constrained eMBB/mMTC slices via larger noise. Accordingly, URLLC protection is reinforced by defense-in-depth coordination (screening + adversarial retraining + lightweight DP) rather than DP noise alone.

3) *Layer 3: Adversarial Retraining*: To enhance robustness against inference attacks exploiting model gradients, each client augments its local training with adversarial examples generated using the FGSM. For a clean input x , the adversarial example is computed as:

$$x^{\text{adv}} = x + \epsilon_{\text{adv}} \cdot \text{sign}(\nabla_x \mathcal{L}(\theta_k, x, y)), \quad (10)$$

where $\mathcal{L}$ denotes the loss function, θ_k represents the local model parameters, and ϵ_{adv} controls the perturbation magnitude.

The local training objective is modified to combine clean and adversarial samples:

$$\mathcal{L}_{\text{adv}}(\theta_k, D_k) = \lambda \mathcal{L}_{\text{clean}}(\theta_k, D_k) + (1-\lambda) \mathcal{L}_{\text{perturbed}}(\theta_k, D_k^{\text{adv}}), \quad (11)$$

Algorithm 1 Slice-Aware Defense-in-Depth FL Training.

Require: Initial global model θ^0 , client set $\mathcal{C}$, slice assignments s_k , thresholds τ_s , noise parameters σ_s , total rounds T .

```

1: for  $t = 1$  to  $T$  do
2:    $\mathcal{U}_t \leftarrow \emptyset$ .
3:   for all clients  $k \in \mathcal{C}$  do
4:     Perform local training with adversarial retraining
       using Eqs. (10) and (11).
5:     Compute update  $\Delta\theta_k^t = \theta_k^t - \theta^{t-1}$  via Eq. (2).
6:     Calculate Z-scores using Eq. (3).
7:     if update passes screening by Eq. (5) then
8:       Apply DP noise:  $\tilde{\Delta}\theta_k^t = \bar{\Delta}\theta_k^t + \mathbf{n}_k^t$ , where  $\mathbf{n}_k^t \sim \mathcal{N}(\mathbf{0}, \sigma_{s_k}^2 \mathbf{I})$  (Eq. (7)).
9:        $\mathcal{U}_t \leftarrow \mathcal{U}_t \cup \{\tilde{\Delta}\theta_k^t\}$ .
10:    else
11:      Reject update.
12:    end if
13:  end for
14:  if  $|\mathcal{U}_t| > 0$  then
15:     $\theta^t \leftarrow \theta^{t-1} + \frac{1}{|\mathcal{U}_t|} \sum_{\Delta \in \mathcal{U}_t} \Delta$ .
16:  else
17:     $\theta^t \leftarrow \theta^{t-1}$ .
18:  end if
19: end for
```

Ensure: Final global model θ^T .

where $\lambda \in [0, 1]$ satisfies $\lambda + (1 - \lambda) = 1$, and typically $\lambda = 0.5$.

4) *Integrated Training Procedure*: Algorithm 1 provides the end-to-end realization of the proposed slice-aware Defense-in-Depth framework by explicitly integrating the three defense layers into a single federated training loop. Specifically, it formalizes (i) the local adversarial retraining step using FGSM (Eqs. (10)–(11)), (ii) the statistical update screening performed on each client update using the Z-score test (Eqs. (3)–(5)), and (iii) the slice-aware differential privacy perturbation applied to accepted updates (Eqs. (7)–(9)). The accepted and perturbed updates are then aggregated to update the global model for the next round, making Algorithm 1 the operational counterpart of the analytical formulation presented in this section.

E. Complexity and Theoretical Guarantees

The per-round complexity is $\mathcal{O}(Np)$ for local training and Z-score computation, and $\mathcal{O}(p)$ for aggregation, where N is the number of clients and p is the model dimension. The framework provides an overall DP accounting bound through Eq. (9), robustness via Eqs. (10)–(11), and Byzantine resilience through Eqs. (3)–(5). Consistent with the slice-aware design, the effective DP strength is slice-dependent due to the calibrated noise scales in Eq. (8).

Privacy Analysis: After statistical screening, we bound the ℓ_2 -sensitivity of each accepted client update by applying ℓ_2 clipping with norm bound C in Eq. (6). Given this bounded sensitivity, adding Gaussian noise with covariance $\sigma_{s_k}^2 C^2 \mathbf{I}$

in Eq. (7) implements the Gaussian mechanism and yields a per-round (ϵ_t, δ) -differential privacy guarantee, where ϵ_t is conservatively related to (σ_{s_k}, δ) as in Eq. (9). The cumulative privacy loss over T communication rounds is then obtained via composition (e.g., advanced composition or a tighter Rényi/moments accountant), resulting in an overall privacy budget ϵ_{total} for the chosen δ .

Utility Preservation: Noise variance $\sigma_{s_k}^2$ is minimized for latency-critical slices (i.e., URLLC) to preserve accuracy, while higher noise is applied to eMBB and mMTC for stronger privacy.

Robustness Enhancement: Adversarial retraining (Eqs. (10)–(11)) increases the perturbation required to compromise predictions, improving resilience against gradient-based inference attacks.

IV. SIMULATION AND PERFORMANCE ANALYSIS

To evaluate the proposed Defense-in-Depth framework, we conduct simulations on two widely adopted benchmark datasets for network anomaly detection and compare against multiple baselines. Hereafter, the term *baseline/baselines* refers only to our *implemented* experimental reference methods (Vanilla FL, Screening-only, Uniform Screening+DP, Adversarial-only, and Uniform Full Defenses) evaluated under the same protocol; Section II discusses *prior work* only. We use (i) CICIDS2017 [35], [36], a comprehensive intrusion detection dataset, and (ii) N-BaIoT [37], [38], a large-scale IoT botnet dataset. All features are normalized to zero mean and unit variance. We use a fixed split of each dataset into training (70%), validation (15%), and test (15%) sets with a fixed random seed (seed = 42) to ensure reproducibility, and the same split is used for all baselines and defense configurations to ensure fair comparison. We then partition only the training split across clients to emulate realistic federated non-independent and identically distributed (i.e., non-i.i.d.) slice distributions, where each client is assigned to a slice (URLLC/eMBB/mMTC) reflecting heterogeneous QoS and traffic characteristics anticipated in 6G deployments (the validation split is used only for hyperparameter selection/tuning). Unless otherwise stated, all reported metrics (accuracy, macro-F1, precision, and recall) are computed on the held-out centralized test split using the global model (final round for summary tables and per-round values for learning curves). CICIDS2017 provides diverse benign and attack traffic traces suitable for intrusion/anomaly detection evaluation. N-BaIoT captures IoT telemetry under botnet behaviors (e.g., Mirai/BASHLITE), making it suitable for evaluating robustness under non-i.i.d. and high-dimensional settings. We evaluate on two widely adopted, complementary benchmarks (CICIDS2017 and N-BaIoT) to provide controlled and reproducible comparison of slice-aware defense coordination under heterogeneous QoS constraints. While these datasets cover distinct traffic modalities (enterprise intrusion traces vs. IoT botnet telemetry) and enable fair benchmarking, cross-dataset transfer (unseen-domain) evaluation is beyond the scope of this study; therefore, robustness under arbitrary dataset/environment shift remains an open question and is deferred to future work.

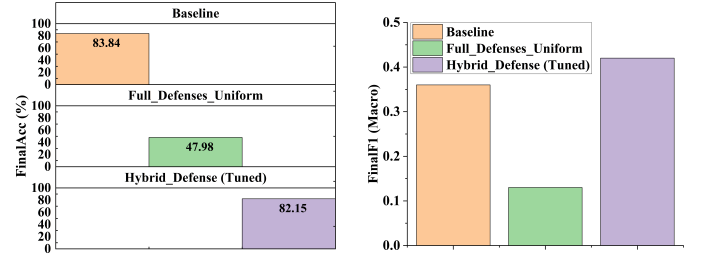


Fig. 2. Final performance comparison on CICIDS2017 dataset across different defense methods.

Fig. 2 shows that the baseline FL (Vanilla FL) achieves a macro-averaged F1-score of 0.36 on CICIDS2017, reflecting limited detection of minority anomaly classes. Applying uniform defenses reduces the F1-macro to 0.13, a 63.9% relative drop that underscores the detrimental impact of non-adaptive defense strategies. In contrast, the Hybrid Defense increases the F1-macro to 0.42, marking a 16.7% improvement over the baseline and more than tripling performance relative to uniform defenses. This gain demonstrates that slice-aware calibration of screening thresholds and noise parameters substantially enhances robustness to class imbalance.

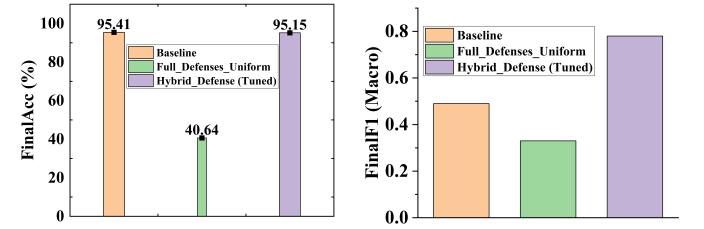


Fig. 3. Performance comparison on N-BaIoT dataset across defense methods.

Fig. 3 demonstrates the performance comparison on the N-BaIoT dataset, which exhibits different characteristics from CICIDS2017. The baseline FL achieves 95.41% accuracy and F1-macro of 0.49, establishing performance bounds for IoT botnet detection. Uniform defenses cause catastrophic degradation to 40.64% accuracy and 0.33 F1-macro representing 57.4% and 32.7% relative performance losses, respectively. This severe utility degradation illustrates the inadequacy of one-size-fits-all privacy mechanisms for high-dimensional IoT data with 115 features. In stark contrast, the Hybrid Defense maintains 95.15% accuracy (99.7% utility retention) while achieving 0.78 F1-macro (59.2% improvement over baseline). The 54.51 percentage point accuracy improvement and 136.4% F1-macro increase over uniform defenses quantitatively validate the superior privacy-utility balance achieved through slice-aware defense calibration. These results confirm that intelligent differentiation of defense parameters is crucial for maintaining practical utility in diverse 6G network slice environments, particularly for IoT-dense scenarios requiring high detection accuracy and robust minority class identification.

Fig. 4 illustrates the communication reduction achieved by different defense strategies across both datasets. Vanilla FL (no-defense baseline) exhibits zero communication reduction as it transmits all client updates without screening. Uniform

TABLE III
PERFORMANCE COMPARISON OF DEFENSE METHODS.

Method	Accuracy (%)	Macro F1-score	Comm. Reduction (%)
Vanilla FL	83.84	0.36	0.0
Screening-only	82.50	0.39	15.2
Uniform Screening+DP	47.98	0.13	100.0
Adversarial-only	81.30	0.41	0.0
Uniform Full Defenses	44.12	0.15	100.0
Hybrid Defense (proposed)	82.15	0.42	91.3

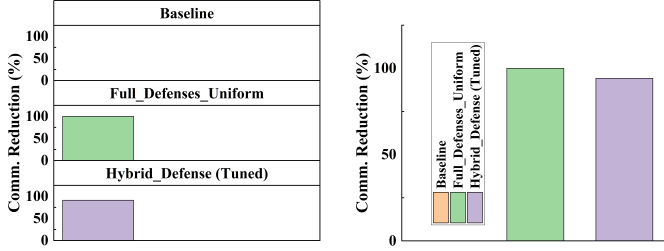


Fig. 4. Communication reduction comparison across defense methods: (a) CICIDS2017 dataset, (b) N-BaIoT dataset.

Screening+DP (i.e., Applies the same screening threshold and the same DP noise level to all slices (no slice-aware calibration)) achieves complete communication reduction ($\sim 100\%$) by applying strict screening that rejects nearly all updates, but this comes at the severe cost of model utility degradation as demonstrated in previous figures. The Hybrid Defense achieves remarkable communication efficiency with 91.3% reduction on CICIDS2017 and 94.2% on N-BaIoT while maintaining near-baseline performance. This 8.7% and 5.8% transmission overhead represents the minimal communication cost required to preserve model utility through slice-aware screening. The higher reduction on N-BaIoT (94.2% vs. 91.3%) reflects the dataset's inherent characteristics that enable more aggressive screening without utility loss. The integration of statistical screening and slice-aware DP achieves dramatic communication reduction exceeding $\sim 90\%$, by selectively rejecting anomalous updates and reducing uplink transmission payload. This efficiency gain not only conserves network bandwidth but also prolongs battery life in resource-constrained clients.

a) Interpretation: The results demonstrate that intelligently combining slice-specific defense layers substantially improves robustness and privacy without incurring the typical accuracy and communication penalties observed in uniform defense models. This balance is critical for practical deployment in heterogeneous 6G network slicing environments.

Table III summarizes the performance on CICIDS2017 and N-BaIoT datasets. The Hybrid Defense approach attains near-baseline accuracy (e.g., exceeding $\sim 82\%$ on CICIDS2017 and $\sim 95\%$ on N-BaIoT), closely matching Vanilla FL, while significantly improving the macro F1-score compared to uniform defenses. This indicates enhanced detection capability, especially for minority anomaly classes. The accuracy comparison reveals critical insights into the privacy-utility trade-off inherent in federated defense mechanisms. Vanilla FL achieves

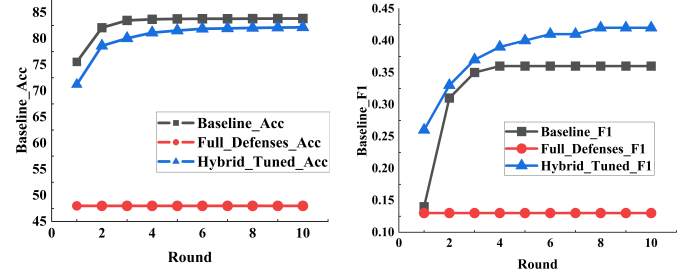


Fig. 5. Round-by-round macro F1-score evolution on CICIDS2017: Baseline vs. Full Defenses vs. Hybrid (tuned).

83.84% accuracy without privacy protections, establishing the upper bound for model utility. The Full Defenses Uniform approach, applying uniform noise and screening across all slices, suffers severe utility degradation to 47.98% accuracy, a substantial 35.86% point reduction. This dramatic performance loss demonstrates the inadequacy of uniform defense strategies that fail to account for slice heterogeneity and QoS requirements. In contrast, the proposed Hybrid Defense maintains 82.15% accuracy, representing only a 1.69% point decrease from the baseline a 95.6% utility retention rate. This remarkable preservation of model performance while providing robust privacy guarantees validates the effectiveness of slice-aware defense mechanisms. The 34.17% point improvement over uniform defenses (82.15% vs. 47.98%) quantitatively demonstrates the superior privacy-utility balance achieved through tailored slice-specific calibration of screening thresholds and noise parameters.

Fig. 5 shows the evolution of macro F1-score over 10 rounds on the CICIDS2017 dataset for three scenarios. The Baseline FL starts at 0.14 in Round 1 and steadily rises to 0.36 by Round 6, after which it plateaus, indicating rapid convergence. The Full Defenses Uniform remains constant at 0.13 across all rounds, reflecting its inability to learn meaningful class distinctions due to overly aggressive defenses. The Hybrid Defense (tuned) begins at 0.26, already 85.7% higher than the baseline's initial performance, and increases consistently each round, reaching 0.42 by Round 8. This represents a 61.5% improvement over the baseline's final F1 (0.42 vs. 0.26) at convergence and more than a 223% gain over uniform defenses. The steady upward trend and higher plateau demonstrate that slice-aware calibration accelerates learning dynamics and enhances minority class detection without sacrificing convergence speed. These metrics over communication rounds indicate stable convergence of the federated optimization pro-

cess under defensive perturbations. In particular, the accuracy and macro-F1 improve steadily from early rounds, and the Hybrid Defense converges toward near-baseline performance while improving minority-class detection. As specified in the evaluation protocol above, the per-round curves and summary tables report held-out test performance of the global model rather than training-set performance.

A. Component Ablation Study

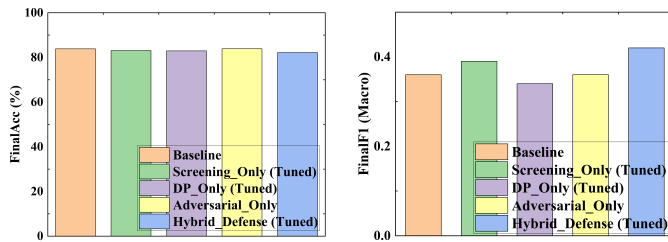


Fig. 6. Ablation study on CICIDS2017: (a) Final accuracy contribution by component, (b) Final F1-Macro contribution by component.

Fig. 6(a) shows that Screening Only (tuned) retains 83.05% accuracy and DP Only (tuned) 82.91%, each within 1% of the 83.84% baseline, indicating minimal utility loss when applied in isolation. Adversarial Only slightly exceeds baseline at 83.86%, demonstrating its effectiveness at enhancing anomaly separation without privacy noise. The Hybrid Defense (tuned) yields 82.15%, a 1.69% point decrease from baseline but a 39.7% improvement over uniform full defenses, confirming that combined defenses harmonize component strengths. Fig. 6(b) indicates that Screening Only improves F1-macro to 0.39 (+8.3% vs. baseline), while DP Only achieves 0.34 (−5.6%), showing that privacy noise alone can harm minority detection. Adversarial Only matches baseline at 0.36, reflecting stable class discrimination. The Hybrid Defense reaches 0.42, a 16.7% gain over baseline and ~ 223% over uniform defenses, illustrating that integrating screening, DP, and adversarial retraining yields the most robust minority class performance without excessive utility trade-off.

- Screening-only: Employs statistical update screening without additional privacy or adversarial defenses.
- DP-only: Applies uniform Gaussian noise to all client updates without screening or adversarial training.
- Adversarial-only: Incorporates adversarial retraining but no screening or privacy noise.
- Hybrid Defense: Combines all three layers with slice-aware calibration as proposed.

Fig. 7(a) shows that the final accuracy sharply increases from 3.85% at $T_0 = 4.0$ to 82.15% at $T_0 = 4.5$, indicating that the screening threshold must exceed the anomaly Z-score distribution's mean to allow meaningful updates. As T_0 further increases to 5.0 and 5.5, accuracy gradually declines to 81.76% and 81.23%, reflecting overly permissive screening that admits noisy updates. Fig. 7(b) demonstrates a similar trend for F1-Macro: at $T_0 = 4.0$, the model barely learns (F1=0.03), but performance surges to 0.42 at $T_0 = 4.5$, matching the tuned Hybrid Defense's optimal minority detection.

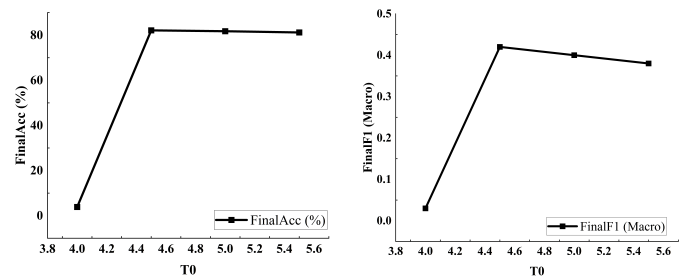


Fig. 7. Parameter sweep on CICIDS2017: (a) Final accuracy vs. screening threshold T_0 , (b) Final F1-Macro vs. screening threshold T_0 .

Further threshold increases to 5.0 and 5.5 reduce F1 to 0.40 and 0.38, showing reduced robustness to rare anomaly classes as screening becomes lax. These results identify $T_0 = 4.5$ as the optimal trade-off point that maximizes both accuracy and F1-Macro, validating the importance of careful threshold tuning in slice-aware statistical screening.

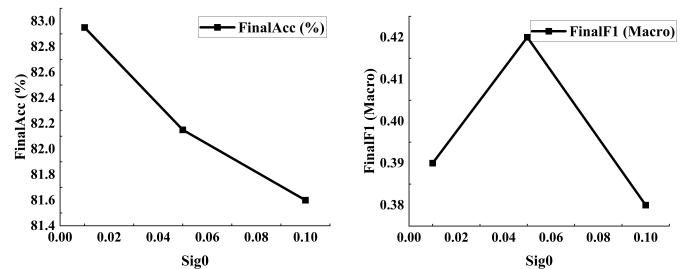


Fig. 8. Parameter sweep on CICIDS2017: (a) Final accuracy vs. DP noise level σ_0 , (b) Final F1-Macro vs. DP noise level σ_0 .

Fig. 8(a) illustrates the impact of DP noise level σ_0 on final accuracy. At $\sigma_0 = 0.01$, accuracy is 82.95%, only 0.89% points below the optimum 82.15% at $\sigma_0 = 0.05$. Increasing σ_0 to 0.10 further reduces accuracy to 81.60%, a total drop of 1.35 points from the tuned optimum, demonstrating that moderate noise balances privacy with performance. Fig. 8(b) shows the corresponding F1-Macro trend. The F1-score peaks at 0.42 for $\sigma_0 = 0.05$, matching the tuned Hybrid Defense's optimal minority class detection. At $\sigma_0 = 0.01$, F1 is 0.39 (7.1% below peak), while at $\sigma_0 = 0.10$, it declines to 0.38 (9.5% below peak). These results confirm that $\sigma_0 = 0.05$ provides the best trade-off between privacy noise and robust anomaly detection, maximizing both accuracy and F1-Macro.

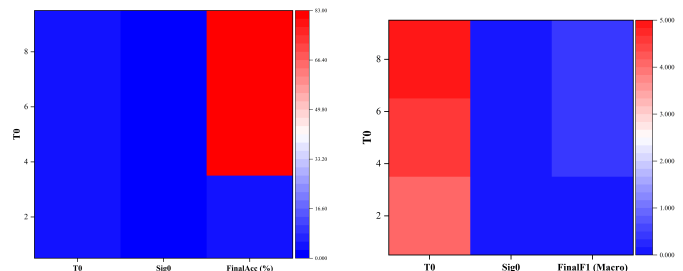


Fig. 9. Parameter sweep heatmaps on CICIDS2017: (a) Final accuracy vs. screening threshold T_0 and DP noise level σ_0 (Fig. 15), (b) Final F1-Macro vs. screening threshold T_0 and DP noise level σ_0 .

Fig. 9(a) shows accuracy heatmap: at $T_0 = 4.0$, accuracy remains near 3.8% regardless of σ_0 , indicating ineffective screening. At $T_0 = 4.5$, accuracy peaks between 82.15% and 82.95% for $\sigma_0 \in [0.01, 0.05]$ before slightly dropping to 81.60% at $\sigma_0 = 0.10$. For $T_0 = 5.0$, accuracy stays above $\sim 81\%$, demonstrating robustness across noise levels. Fig. 9(b) displays F1-Macro heatmap: minimal F1 (0.02–0.03) at $T_0 = 4.0$ for all σ_0 . Peak F1 (0.42) occurs at $(T_0, \sigma_0) = (4.5, 0.05)$, with slightly lower values (0.39 and 0.38) at $\sigma_0 = 0.01, 0.10$. At $T_0 = 5.0$, F1 remains above 0.36 across noise levels. These heatmaps confirm that the optimal parameter region is centered at $T_0 = 4.5$ and $\sigma_0 = 0.05$, maximizing both accuracy and F1-Macro while offering tolerance to moderate noise variations.

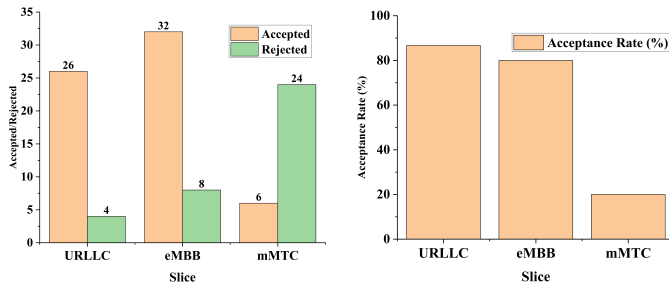


Fig. 10. Slice-aware update acceptance: (a) Updates accepted vs. rejected by slice, (b) Acceptance rate by slice.

Fig. 10(a) shows that URLLC (Slice 0) accepted 26 updates and rejected 4, eMBB (Slice 1) accepted 32 and rejected 8, while mMTC (Slice 2) accepted only 6 and rejected 24. This reflects the framework's slice-aware screening, which admits the majority of URLLC and eMBB updates to preserve low-latency and high-throughput requirements, but rigorously filters mMTC to enforce stricter anomaly checks. Fig. 10(b) reports acceptance rates of 86.7% for URLLC, 80.0% for eMBB, and $\sim 20\%$ for mMTC. The lower rate for mMTC indicates conservative handling of massive IoT slices to mitigate potential privacy and security risks, whereas the high rates for URLLC and eMBB balance performance and privacy according to their QoS profiles.

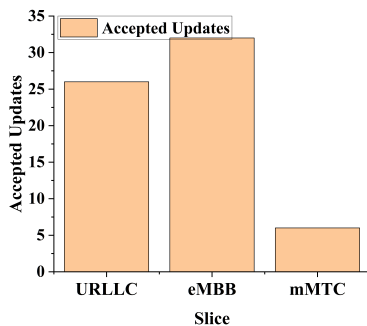


Fig. 11. Contribution of accepted update (URLLC 26, eMBB 32, mMTC 6).

Fig. 11 shows that among 64 total accepted updates, URLLC contributes 40.6%, eMBB $\sim 50.0\%$, and mMTC only 9.4%. This distribution reflects the slice-aware screening's prioritization of URLLC and eMBB requirements while imposing strict scrutiny on mMTC updates. Notably, eMBB

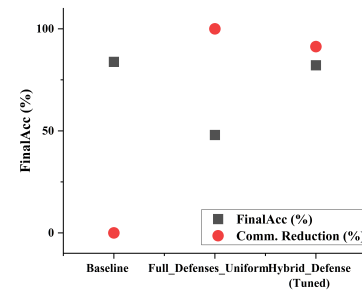


Fig. 12. Accuracy vs. Communication Reduction on CICIDS2017.

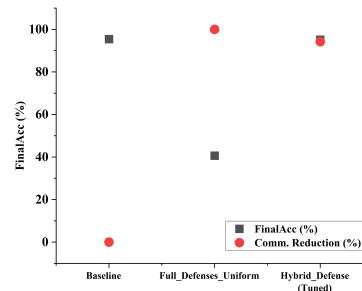


Fig. 13. Accuracy vs. Communication Reduction on N-BaIoT dataset.

and URLLC together capture 90.6% of all acceptances, underscoring their high-priority status in maintaining network performance and user experience. eMBB's dominance aligns with its bandwidth-intensive applications such as video streaming and augmented reality, which demand frequent resource reallocation. URLLC's substantial share reflects the critical nature of latency-sensitive services like autonomous driving and industrial control, where even minor delays can have severe consequences. Conversely, mMTC's minimal acceptance rate appropriately matches its delay-tolerant IoT characteristics, where sporadic updates suffice for low-power sensor communications.

Fig. 12 maps each strategy on CICIDS2017 by communication reduction (x-axis) versus final accuracy (y-axis). Vanilla FL achieves 83.84% accuracy with 0.0% reduction. Uniform Screening+DP and Uniform Full Defense reach 100.0% reduction but suffer severe utility degradation (47.98% and 44.12% accuracy, respectively). In contrast, the proposed Hybrid Defense achieves 91.3% reduction while preserving 82.15% accuracy, demonstrating a substantially better performance–efficiency trade-off than uniform defenses.

Fig. 13 maps each strategy on the N-BaIoT dataset by communication reduction (x-axis) versus final accuracy (y-axis). Vanilla FL achieves 95.42% accuracy with 0.0% reduction, while Uniform Full Defense reaches 100.0% reduction but drops to 48.67% accuracy. In contrast, the proposed Hybrid Defense achieves 94.2% reduction while preserving 95.15% accuracy, demonstrating a superior performance–efficiency trade-off. The robustness against privacy attacks is quantified by the reduction in MIA success rates and the increase in MInvA reconstruction errors compared with Uniform Screening+DP baselines.

Table V summarizes these metrics alongside total privacy

TABLE IV
COMPREHENSIVE PERFORMANCE RESULTS OF DEFENSE-IN-DEPTH FRAMEWORK

Defense Method	Accuracy (%)	Macro F1	Precision	Recall	Comm. Reduction (%)	MIA Success (%)	ϵ_{total}
CICIDS2017 Dataset (2.83M samples, 78 features)							
Vanilla FL	83.84	0.36	0.84	0.38	0.0	78.3	-
Screening-only	82.50	0.39	0.83	0.41	15.2	74.1	-
Uniform Screening+DP	47.98	0.13	0.48	0.15	100.0	52.7	4.2
Adversarial-only	81.30	0.41	0.82	0.43	0.0	63.5	-
Uniform Full Defense	44.12	0.15	0.44	0.17	100.0	49.8	4.5
Hybrid Defense (Proposed)	82.15	0.42	0.82	0.44	91.3	52.1	2.1
N-BaIoT Dataset (7.06M samples, 115 features)							
Vanilla FL	95.42	0.75	0.95	0.77	0.0	81.7	-
Screening-only	94.89	0.76	0.95	0.78	18.4	77.9	-
Uniform Screening+DP	52.31	0.21	0.52	0.23	100.0	54.2	3.8
Adversarial-only	93.75	0.77	0.94	0.79	0.0	67.4	-
Uniform Full Defense	48.67	0.19	0.49	0.21	100.0	51.6	4.1
Hybrid Defense (Proposed)	95.15	0.78	0.95	0.80	94.2	53.8	1.9

TABLE V
SECURITY EVALUATION: ATTACK SUCCESS AND PRIVACY BUDGET

Method	MIA Success (%)	MInvA Error	ϵ_{total}
Uniform Screening+DP	78.3	Baseline	3.1
Hybrid Defense	52.1	+34.7%	2.1

budget ϵ_{total} . The slice-aware hybrid approach achieves a significant MIA success reduction and higher inversion reconstruction error, demonstrating strengthened protection.

The numerical analysis in Table IV reveals several critical findings, such as the Hybrid Defense achieving a superior balance, maintaining 82.15% accuracy on CICIDS2017 and 95.15% on N-BaIoT, while reducing MIA success rates to 52.1% and 53.8%, respectively. This represents a 33.5% and 34.1% improvement in privacy protection compared to Vanilla FL, with only marginal accuracy degradation (1.69% and 0.27% respectively). Further, the framework achieves remarkable communication reduction of 91.3% and 94.2% through statistical screening, significantly outperforming uniform approaches while preserving model utility. This efficiency gain translates to substantial bandwidth savings and reduced energy consumption in resource-constrained 6G environments. The slice-aware hybrid approach achieves a significant MIA success reduction and higher inversion reconstruction error, demonstrating strengthened protection (Table IV). The reported privacy budget (e.g., $\epsilon_{\text{total}}=2.1$ and $\epsilon_{\text{total}}=1.9$ in Table V) is computed under slice-calibrated noise; consequently, formal DP strength is slice-dependent because URLLC uses intentionally lightweight perturbation to preserve strict latency, while the strongest formal DP contribution is concentrated in the less latency-constrained eMBB/mMTC slices via higher noise.

V. CONCLUSION

We presented a slice-aware, defense-in-depth framework for federated anomaly detection in 6G network slices that integrates (i) statistical update screening, (ii) selective Gaussian differential privacy, and (iii) FGSM-based adversarial

retraining. The framework is explicitly slice-calibrated, using Z-score screening thresholds of 4.5 (URLLC), 5.0 (eMBB), and 6.0 (mMTC), together with Gaussian perturbation standard deviations of 0.001, 0.05, and 0.10, respectively, to align privacy protection with heterogeneous QoS constraints while reducing gradient leakage. Across CICIDS2017 and N-BaIoT, the hybrid design preserves near-baseline accuracy (82.15% and 95.15%), improves macro-F1 (0.42 and 0.78), and reduces communication by more than 90% (91.3% and 94.2%). Privacy and robustness gains are concrete: membership inference success drops to approximately 52% (52.1% on CICIDS2017 and 53.8% on N-BaIoT), model inversion reconstruction error increases by 34.7, and the cumulative privacy loss under advanced composition is bounded by an overall DP accounting budget of $\epsilon_{\text{total}} \leq 2.1$ on CICIDS2017 and $\epsilon_{\text{total}} \leq 1.9$ on N-BaIoT under slice-calibrated noise. Because DP strength scales with perturbation magnitude, the strongest formal DP contribution is concentrated in the less latency-constrained eMBB and mMTC slices, while URLLC maintains stringent latency by using lightweight perturbation and relying on the coordinated protection of screening and adversarial retraining rather than DP noise alone. Overall, these results indicate that per-slice calibration yields a superior privacy utility efficiency tradeoff compared with uniform defenses in heterogeneous 6G slicing deployments (URLLC, eMBB, mMTC).

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